

elegant-polimi-thesis documentation

This chapter is the result of the documentation generated from the source code. Since it's not strictly needed to use the template – most of the functions are straightforward – it's quite barebones.

This document contains them all, even those that are not exposed to the user, which are marked with “_” before.

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Theses templates

polimi-thesis

The main thesis formatting function.

Parameters

```
polimi-thesis(  
  title: str,  
  author: str,  
  supervisor: str array,  
  cosupervisor: str array,  
  academic-year: str,  
  tutor: str,  
  cycle: str,  
  chair: str,  
  student-id: str,  
  course: str,  
  frontispiece: "phd" "deib-phd" "cs-eng-master" "classical-master",  
  custom-logo: image,  
  body  
) -> content
```

title str

Title of the thesis.

Default: "Thesis Title"

author str

Author of the thesis.

Default: none

supervisor str or array

Supervisor(s) of the thesis.

Default: none

cosupervisor str or array

Cosupervisor(s) of the thesis.

Default: none

academic-year str

Academic year of the thesis. If empty, defaults to "2025-2026".

Default: ""

tutor str

Tutor of the thesis.

Default: none

cycle `str`

Cycle of the thesis.

Default: `none`

chair `str`

Chair of the thesis.

Default: `none`

student-id `str`

Student ID.

Default: `none`

course `str`

Student course.

Default: `none`

frontispiece `"phd" or "deib-phd" or "cs-eng-master" or "classical-master"`

Frontispiece of the thesis. Can be either: phd, deib-phd, cs-eng-master or classical-master.

Default: `"phd"`

custom-logo `image`

Custom logo of the thesis.

Default: `none`

polimi-article-format-thesis

The thesis article format styling function.

Parameters

```
polimi-article-format-thesis(  
  title: str,  
  author: str,  
  supervisor: str,  
  cosupervisor: str array,  
  academic-year: str,  
  student-id: str,  
  course: str,  
  abstract: content,  
  keywords: str,  
  logo: path,  
  body  
) -> content
```

title str

Title of the thesis.

Default: "Thesis Title"

author str

Author of the thesis.

Default: none

supervisor str

supervisor of the thesis.

Default: none

cosupervisor str or array

cosupervisor(s) of the thesis.

Default: none

academic-year str

Academic year of the thesis. If empty, defaults to "2025-2026".

Default: ""

student-id str

Student ID.

Default: none

course `str`

Student course.

Default: `none`

abstract `content`

Abstract.

Default: `[]`

keywords `str`

Keywords, that appear below the abstract (and in the PDF metadata).

Default: `none`

logo `path`

Logo of the thesis.

Default: `image("img/logo_ingegneria.svg", width: 83mm)`

polimi-executive-summary

The executive summary styling function.

Parameters

```
polimi-executive-summary(  
  title: str,  
  author: str,  
  supervisor: str,  
  cosupervisor: str array,  
  academic-year: str,  
  course: str,  
  logo: path,  
  body  
) -> content
```

title `str`

Title of the thesis.

Default: `"Thesis Title"`

author `str`

Author of the thesis.

Default: `none`

supervisor `str`

supervisor of the thesis.

Default: `none`

cosupervisor `str` or `array`

cosupervisor(s) of the thesis.

Default: `none`

academic-year `str`

Academic year of the thesis. If empty, defaults to “2025-2026”.

Default: `""`

course `str`

Student course.

Default: `none`

logo `path`

Logo of the thesis.

Default: `image("img/logo_ingegneria.svg", width: 83mm)`

_blank-toc

Helper function to manually add blank space above an outline entry (similar to $\text{\LaTeX}$'s `\addtocontents{toc}{\vspace{1em}}`).

Parameters

`_blank-toc(space: length) -> content`

space `length`

Vertical space to add before the next outline entry.

Default: `1em`

frontmatter

Frontmatter section. Similar to $\text{\LaTeX}$'s `\frontmatter`, it is meant to be only used in the thesis. It sets the page numbering to "i" and numbering: `none` for headings.

Parameters

`frontmatter`(body) -> `content`

acknowledgements

Acknowledgements section. It sets numbering: `none` for headings.

Parameters

`acknowledgements`(body) -> `content`

mainmatter

Mainmatter section. Similar to $\text{\LaTeX}$'s `\mainmatter`, it is meant to be only used in the thesis. It sets to page numbering to "1", heading numbering to "`1.1`" + `h()` and resets the page counter.

Parameters

`mainmatter`(body) -> `content`

appendix

Appendix section. Similar to $\text{\LaTeX}$'s `\appendix`. It sets heading numbering to "`A.1`" and resets their counter.

Parameters

`appendix`(body) -> `content`

backmatter

Backmatter section. Similar to $\text{\LaTeX}$'s `\backmatter`, it is meant to be only used in the thesis. It sets heading numbering to `none`.

Parameters

`backmatter`(body) -> `content`

_lists-entries-style

Internal helper function to create the custom lists of figures and table.

Parameters

```
_lists-entries-style(  
  outline-entry: outline-entry,  
  kind: function  
) -> content
```

outline-entry `outline-entry`

Outline entry to edit.

kind `function`

The kind of the outline entry element (image or table).

nomenclature

Displays a simple nomenclature with keys and values.

```
#nomenclature(  
  "Polimi": "Politecnico di Milano",  
  "CdL": "Corso di Laurea",  
  "CCS": "Consigli di Corsi di Studio",  
  "CFU": "Crediti Formativi  
Universitari",  
)
```

Nomenclature

<i>POLIMI</i>	Politecnico di Milano
<i>CDL</i>	Corso di Laurea
<i>CCS</i>	Consigli di Corsi di Studio
<i>CFU</i>	Crediti Formativi Universitari

Parameters

```
nomenclature(  
  dict: dictionary,  
  indented: bool  
) -> content
```

dict `dictionary`

Dictionary that hold keys and values.

indented `bool`

Whether to indent or not the nomenclature.

Default: `true`

theorems-init

Utility function to initialize the theorem environments¹:

¹Provided by great-theorems package.

- theorem
- proposition
- lemma
- remark

Parameters

`theorems-init`(body) -> `content`

subfigure

Creates a new subfigure with the given arguments and an optional label. From hallon package.

Parameters

```
subfigure(
  ..args,
  outlined: bool,
  label: str,
  body
) -> content
```

outlined `bool`

Whether to outline this subfigure or not.

Default: `false`

label `str`

Unique label to reference this subfigure

Default: `none`

toc `content`

Custom-built `outline()`.

list-of-figures `content`

List of figures. Similar to $\text{\LaTeX}$'s `\listoffigures`.

list-of-tables `content`

List of tables. Similar to $\text{\LaTeX}$'s `\listoftables`.

theorem `content`

Theorem block.

proposition `content`

Proposition block.

lemma `content`

Lemma block.

remark `content`

Remark block.

proof `content`

Proof block.

Utilities

`_empty-page`

Adds an empty page between an odd page and the next. Used to check when to remove the header and place a raggiera in the bottom left corner.

Parameters

```
_empty-page() -> content
```

`_is-page-between-labels`

Check if the current page is between the given labels.

Parameters

```
_is-page-between-labels(  
  start-label: label,  
  end-label: label,  
  function: function  
) -> bool
```

start-label `label`

The start label.

end-label `label`

The end label.

function `function`

Function to apply to the array.

`_is-page-empty`

Check if a page is empty.

Parameters

`_is-page-empty()` -> `bool`

`_is-page-in-outline`

Check if a page is in an outline.

Parameters

`_is-page-in-outline()` -> `bool`

`_raggiera-image`

Inserts a raggiera, given a specified width.

Parameters

`_raggiera-image(side: length)` -> `content`

side `length`

Side of the raggiera.

`banner`

Draw a banner.

Parameters

`banner(body: content)` -> `content`

body `content`

Body of the banner.

tab-numbering

Alternative numbering: "1.1." + h-space.

Parameters

```
tab-numbering(  
  h-space: length ,  
  ..n: arguments  
) -> content
```

h-space `length`

Horizontal space.

Default: `0.7em`

..n `arguments`

Numbers.

chapter-numbering

Chapter numbering.

Parameters

```
chapter-numbering(..n: arguments) -> content
```

..n `arguments`

Numbers.

header-numbering

Numbering used in the theses header.

Parameters

```
header-numbering(..n: arguments) -> content
```

..n `arguments`

Numbers.

_show-field

Helper function to detect whether a field is present and, if true, show it.

Parameters

```
_show-field(  
    prefix: str,  
    field: variable,  
    separator: function content  
) -> content
```

prefix `str`

Prefix (e.g. "Prof: ").

field `variable`

Exact field to check (e.g. title).

separator `function` or `content`

Separator between fields.

Default: `linebreak()`

_show-starvisor

Helper function to handle (co)supervisor(s).

Parameters

```
_show-starvisor(  
    starvisor: str array ,  
    locale-key: "supervisor" "cosupervisor" ,  
    separator: string content ,  
    join-separator: string content ,  
    key: function ,  
    out: function  
) -> content
```

starvisor `str` or `array`

(Co)Supervisor(s) of the thesis.

locale-key `"supervisor"` or `"cosupervisor"`

The key to look for in the locale dictionary

separator `string` or `content`

How to separate the key from the content.

Default: `": "`

join-separator `string` or `content`

How to separate multiple starvisors.

Default: `", "`

key `function`

Function that will be applied to the key.

Default: `x => x`

out `function`

Function that will be applied to the content.

Default: `x => x`

`_show-author`

Helper function to show localized author.

Parameters

`_show-author()` -> `content`

`_show-academic-year`

Helper function to show localized academic year.

Parameters

`_show-academic-year(year)` -> `content`

`_show-cycle`

Helper function to show the thesis cycle.

Parameters

`_show-cycle(cycle)` -> `content`

_spacer

Empty block.

Parameters

```
_spacer(  
    width,  
    height  
) -> content
```

_stroke-no-border

Internal stroke.

Parameters

```
_stroke-no-border(color) -> stroke
```

_poli-header

Custom made header.

Parameters

```
_poli-header(  
    text-args: dict,  
    self  
) -> content
```

text-args dict

The text arguments for the title.

Default: (size: 15pt, weight: "regular", fill: rgb("#5e5e5e"))

_poli-bg-split

Apply a fill colour to the panes.

Parameters

```
_poli-bg-split() -> content
```

_divider-body

New section body, that is title, lettering and chapter number.

Parameters

```
_divider-body(  
  title,  
  lettering,  
  number,  
  text-fill  
) -> content
```

_divider-bg-args

Style the new section background.

Parameters

```
_divider-bg-args(  
  fill,  
  stroke  
) -> content
```

_basic-subslide-preamble

Standard subslide preamble.

Parameters

```
_basic-subslide-preamble(  
  self,  
  inset,  
  text-args  
) -> content
```

_sizes dict

TeX sizes to match original templates (<https://tex.stackexchange.com/questions/24599/what-point-pt-font-size-are-large-etc>)

_document-state state

The current state (title page, *matter...)

_document-type state

The current type of document.

_localization dict

Localization dictionary. It could be deprecated since theses are written exclusively in English.

bluepoli color

Signature PoliMi colour (■), used in headings and labels.

_pane dict

The background panes used in some slides.

_numbering-with-padding numbering

(1..20) -> (01,02,03,..., 10, 11, 12,...,20)

Figures styling

_subfigure-caption

It displays the caption of subfigures. Use “(a)” that will be colored based on colored-headings parameter.

Parameters

```
_subfigure-caption(  
    colored-caption: bool,  
    parent: figure,  
    it  
) -> content
```

colored-caption bool

Whether to apply bluepoli on captions

Default: true

parent figure

Parent figure of this subfigure

Default: none

`_style-figures`

Style-figures handles (optional heading-dependent) numbering of figures and subfigures. Can modify the caption of all figures.

Parameters

```
_style-figures(  
    colored-caption: bool,  
    heading-levels: int,  
    body  
) -> none
```

colored-caption `bool`

Whether to apply bluepoli on captions ([Figure 1](#)).

Default: `true`

heading-levels `int`

Heading level to account for in the numbering.

Default: `0`

Presentation

poli-slide

Basic slide.

Parameters

```
poli-slide(  
    config: dictionary,  
    repeat: auto bool,  
    setting,  
    composer,  
    ..args,  
    body  
) -> content
```

config dictionary

Touying overrides for this slide.

Default: (:)

repeat auto or bool

Whether to repeat this slide.

Default: auto

title-slide

The title slide. The style can be specified via the corresponding argument. For the background and logos style you must also specify a background and logos.

Parameters

```
title-slide(  
    style: "default" "white" "logos" "background",  
    background: image,  
    logos: array,  
    config: dictionary  
) -> content
```

style "default" or "white" or "logos" or "background"

Style of the title slide.

Default: "default"

background image

Required background for the “background” style; otherwise ignored.

Default: none

logos array

Required logos for the “logos” style; otherwise ignored.

Default: ()

config dictionary

Touying overrides for this slide.

Default: (:)

make-outline

Generate the outline based on headings.

Parameters

```
make-outline(config: dictionary) -> content
```

config dictionary

Touying overrides for this slide.

Default: (:)

outline-slide

Generate the outline based on headings.

Parameters

```
outline-slide(..args) -> content
```

exact-split-slide

Slide exactly split in 2 or 4. Requires content for each split.

Parameters

```
exact-split-slide(  
    splits: array,  
    config: dictionary  
) -> content
```

splits array

Content of each split. Usually a title with a small description.

Default: ()

config dictionary

Touying overrides for this slide.

Default: (:)

split-slide

Split a slide in two panes – similar to header spacing.

Parameters

```
split-slide(  
  left: content,  
  right: content,  
  config: dictionary  
) -> content
```

left content

Left split content.

Default: []

right content

Right split content.

Default: []

config dictionary

Touying overrides for this slide.

Default: (:)

plot-slide

Slide with coloured background. Used for plots.

Parameters

```
plot-slide(  
  config: dictionary,  
  body  
) -> content
```

config dictionary

Touying overrides for this slide.

Default: (:)

image-slide

Slide with image on the left, right or as whole background. Deprecated. Use [background-slide\(\)](#).

Parameters

```
image-slide(  
  image: image ,  
  style: "background" or "left" ,  
  config: dictionary ,  
  body  
) -> content
```

image image

Image for the slide.

Default: **none**

style "background" or "left"

Style of the image, that is where it will be placed and sized.

Default: **"background"**

config dictionary

Touying overrides for this slide.

Default: (:)

background-slide

Slide with image as whole background.

Parameters

```
background-slide(  
  image: image ,  
  config: dictionary  
) -> content
```

image image

Image for the slide.

Default: **none**

config dictionary

Touying overrides for this slide.

Default: (:)

focus-slide

Last slide. Usually it contains “Thanks for listening!” or similar phrases.

Parameters

```
focus-slide(  
  lettering: content,  
  config: dictionary,  
  body  
) -> content
```

lettering content

Bottom lettering.

Default: **none**

config dictionary

Touying overrides for this slide.

Default: (:)

_new-section-slide

The so-called “Divisorio”, a new section. Invoked with a level 1 heading.

Parameters

```
_new-section-slide(  
  config,  
  ..args  
) -> content
```

`_new-subsection-slide`

The other so-called “Divisorio”, but for lvl.2 heading and in a light background. Unused.

Parameters

```
_new-subsection-slide(  
  config,  
  ..args  
) -> content
```

`polimi-digital-presentation`

Main presentation function.

Parameters

```
polimi-digital-presentation(  
  divider-style: "dark" "light",  
  subtitle-every-style: bool,  
  ..args,  
  body  
) -> content
```

divider-style "dark" or "light"

The style of the divider.

Default: "dark"

subtitle-every-style bool

Whether to use the subtitle on all title styles.

Default: false